American Computer Science League

2024-2025 ● Contest 2: Short Problems ● Senior Division

1. Prefix/Infix/Postfix Notation

Translate the following postfix expression to prefix:

a b / c / d 2 ^ e / −

A. − / / a b c ^ / d 2 e
B. / / − a b c / ^ d 2 e
C. − / / a b / c ^ d 2 e
D. − / / a b c / ^ d 2 e
E. / − / a b c / ^ d 2 e

2. Prefix/Infix/Postfix Notation

Define a # b as follows: a # b = max(a,b)
Evaluate the following postfix expression if all numbers are single digits:

4 2 * 4 3 + # 2 / 5 9 − 1 # +

A. 0
B. 3
C. 5
D. 7
E. 8

3. Bit-String Flicking

Evaluate:

(NOT 11010) OR (10111 AND 10001) XOR 10101

A. 00101
B. 00100
C. 10111
D. 10101
E. 00111

4. Bit-String Flicking

Solve the following equation for X if X is a 5-bit string:

(LCIRC-1 (10110 OR 01110)) AND X = 10001

A. 00001
B. *1111
C. 100*1
D. 10101
E. 110*0

5. LISP

Evaluate the following LISP expression:

(ADD (EXP 2 4) (SUB 8 3) (MULT 2 1 3 4))

A. 24
B. 29
C. 37
D. 40
E. 45

6. LISP

Evaluate the following LISP expression:

(CDR (CAR (CDR (CDR '((a (b c)) d (e (f g) h) j)))))

A. ((f g) h)
B. (a (b c))
C. (e (f g) h)
D. (b c)
E. (d)